

BANGKOK METROPOLIS, Thailand

WMO: 484550

Lat: 13.7264N

Lon: 100.5600E

Elev: 4

StdP: 101.27

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 20.4	(c) 21.9	(d) 12.2	(e) 8.8	(f) 24.2	(g) 13.9	(h) 9.9	(i) 24.9	(j) 4.4	(k) 29.2	(l) 4.0	(m) 29.0	(n) 0.6	(o) 60	(p) 0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 7.1	(c) 36.2	(d) 26.9	(e) 35.5	(f) 26.8	(g) 34.8	(h) 26.6	(i) 28.4	(j) 33.1	(k) 28.0	(l) 32.6	(m) 27.7	(n) 32.2	(o) 2.0	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.3	23.1	30.7	26.9	22.6	30.3	26.6	22.2	30.1	91.8	33.3	89.9	32.8	88.4	32.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.6	(b) 4.0	(c) 3.4	(e) 18.1	(f) 38.0	(g) 2.0	(h) 0.9	(i) 16.6	(j) 38.7	(k) 15.4	(l) 39.2	(m) 14.3	(n) 39.7	(o) 12.9	(p) 40.4		
(a) 4.6	(b) 4.0	(c) 3.4	(e) 14.4	(f) 29.4	(g) 2.0	(h) 1.1	(i) 13.0	(j) 30.1	(k) 11.8	(l) 30.8	(m) 10.7	(n) 31.4	(o) 9.2	(p) 32.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.4	27.8	29.1	30.3	31.4	30.7	30.0	29.5	29.4	29.1	29.0	28.8	27.6
	DBStd	1.80	1.75	1.52	1.54	1.48	1.57	1.15	1.10	1.02	1.08	1.19	1.55	2.09
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7079	552	536	629	641	641	601	606	602	572	589	563	546
	CDD18.3	4037	294	302	371	391	383	351	348	344	322	331	313	288
	CDH23.3	49900	3197	3595	4793	5383	5152	4586	4358	4241	3787	3914	3731	3164
	CDH26.7	23054	1259	1578	2411	3027	2721	2237	1951	1848	1546	1613	1591	1273
Wind	WSAvg	1.0	0.8	1.2	1.4	1.3	1.1	1.1	1.1	1.1	0.9	0.7	0.8	0.8
Precipitation	PrecAvg	1545	16	24	38	79	216	159	165	211	342	267	53	12
	PrecMax	2240	131	125	176	360	554	412	349	575	676	561	294	103
	PrecMin	876	0	0	0	0	30	28	32	50	165	22	0	0
	PrecStd	306	27	31	44	76	126	86	74	106	110	113	56	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	35.0	36.2	37.4	37.3	35.8	35.0	34.8	34.7	34.5	35.0	34.5
		MCWB	24.3	24.6	25.5	26.7	27.4	27.1	26.7	26.8	26.7	25.9	25.5	25.1
	2%	DB	33.3	34.1	35.2	36.4	36.1	34.6	34.0	33.8	33.6	33.7	34.0	33.5
		MCWB	24.3	25.1	26.0	27.0	27.4	26.8	26.6	26.6	26.6	25.9	25.3	24.7
	5%	DB	32.6	33.3	34.4	35.6	35.1	33.8	33.2	33.0	32.8	33.0	33.1	32.7
		MCWB	24.2	25.2	26.2	27.1	27.2	26.8	26.5	26.4	26.4	25.9	25.2	24.4
	10%	DB	31.8	32.7	33.7	34.9	34.1	33.0	32.4	32.2	32.1	32.2	32.3	31.8
		MCWB	24.0	25.2	26.2	27.0	27.0	26.6	26.2	26.2	26.2	25.9	25.0	24.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.1	27.7	28.0	29.2	29.0	28.2	28.2	28.4	27.7	27.6	27.6	26.8
		MCDB	31.1	32.4	33.6	34.5	33.9	32.7	32.5	32.2	31.9	31.8	31.8	31.3
	2%	WB	26.4	27.0	27.6	28.5	28.3	27.7	27.5	27.5	27.3	27.2	27.0	26.2
		MCDB	30.2	31.5	32.6	33.4	32.9	32.1	31.7	31.4	31.3	31.1	31.0	30.6
	5%	WB	25.9	26.6	27.2	28.0	27.9	27.5	27.1	27.1	27.0	26.8	26.6	25.8
		MCDB	29.7	30.8	31.8	32.8	32.4	31.7	31.3	31.0	31.0	30.6	30.5	30.1
	10%	WB	25.5	26.2	26.9	27.7	27.6	27.2	26.7	26.7	26.7	26.6	26.2	25.3
		MCDB	29.2	30.2	31.3	32.3	32.1	31.3	30.8	30.6	30.5	30.2	30.0	29.7
Mean Daily Temperature Range	5% DB	MDBR	8.1	7.4	6.9	7.1	6.7	6.2	5.8	5.9	6.2	6.4	6.9	7.7
		MCDBR	8.6	7.9	7.6	7.6	7.5	7.1	6.8	6.8	7.2	7.3	7.6	8.0
		MCWBR	2.7	2.3	2.1	2.2	2.3	2.4	2.4	2.4	2.4	2.1	2.3	2.5
	5% WB	MCDBR	6.9	6.9	6.9	7.0	7.1	6.6	6.3	6.4	6.6	6.4	6.8	7.0
		MCWBR	2.5	2.1	2.0	2.2	2.4	2.5	2.5	2.5	2.4	2.3	2.3	2.4
Clear-Sky Solar Irradiance	taub	0.545	0.612	0.638	0.576	0.499	0.466	0.458	0.460	0.484	0.580	0.526	0.511	
	taud	1.957	1.825	1.795	1.984	2.255	2.346	2.367	2.349	2.271	1.962	2.090	2.077	
	Ebn at Noon	755	724	718	761	808	827	835	844	828	740	768	774	
	Edh at Noon	181	214	225	186	140	126	124	128	138	184	157	157	
All-Sky Solar Radiation	RadAvg	5.10	5.54	5.69	6.09	5.56	5.30	5.06	5.11	4.94	4.79	4.93	4.91	
	RadStd	0.29	0.23	0.42	0.38	0.39	0.28	0.26	0.32	0.28	0.26	0.34	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.45	N/A	+0.33	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	+0.20	+0.22	N/A	N/A	N/A	N/A

Nomenclature: See separate page